CS 494: Security for and with GenAI-Based Systems
Course Project: Proposal Guidelines

Proposal due: Monday, October 26, 2026. A one- to two-page proposal description is sufficient. Projects may be completed individually or in teams of two or three; working in teams is encouraged.

Project topics. Your project should investigate a problem in one of the following broad areas:

- **Security of generative AI systems.** Examples include attacks and defenses for AI agents, new prompt-injection vectors, techniques for preventing prompt injection, and related security problems in GenAI-based applications.
- **Software security using generative AI.** Examples include vulnerability analysis of an open-source software system, secure software design and development, and techniques for identifying and fixing security bugs with the assistance of GenAI tools.

These examples are illustrative rather than exhaustive. You are welcome to propose other topics that fit the scope of the course.

Expected effort and use of GenAI. The project should represent approximately **15–20 hours of work per student**. (Note: this expectation excludes the effort for writing the report and preparing the presentation.) You may use GenAI tools, but you must document how you used them and where your own intellectual and technical effort went. For team projects, identify the specific roles and responsibilities of each member. Your final report should make these contributions clear.

What to include in the proposal. Briefly address the following:

- **Title and team members:** Give the project a descriptive title and list all team members.
- **Project description:** Explain the problem you plan to study, your objective, and the approach you expect to take.
- **Background:** Summarize any relevant prior work, tools, systems, or other context needed to understand the project.
- **Timeline and work plan:** Describe the main steps you will complete between the proposal and final submission. For team projects, indicate who will be responsible for each part.
- **Deliverables:** Specify what you expect to submit, such as code, experiments, a demonstration, or an analysis, together with a **brief final report (3 pages or less)** as well a presentation describing your work and results.

Project presentations. Projects will be presented in class on **Monday, November 30**, and **Wednesday, December 2, 2026**. Plan your work so that you can present your approach and results on these dates.